

# TRISTAN PITT

## Graduate Software Engineer



Portfolio



LinkedIn



GitHub



TristanPitt01@gmail.com

## Professional Summery

Dedicated and technically proficient Graduate Games Programmer with a First-Class Honours degree in Computer Science. Strong foundational knowledge in **C++**, **C#**, and **Python**, with hands-on experience developing 2D and 3D titles in **Unreal Engine 5** and **Unity**. Proven ability to debug complex systems and optimize game performance. A collaborative team player with a passion for creating immersive gameplay experiences and a commitment to writing clean, scalable code.

## PREVIOUS EMPLOYMENT

### Northrak

Game Programmer Intern

(Feb 2025 – Present)

I play a versatile role across the full development lifecycle, responsible for architecting and implementing core gameplay systems using Unreal Engine. My work spans the entire technical stack, from engineering responsive UI frameworks and player interfaces to fine-tuning gameplay balancing for an optimal user experience. I am heavily involved in performance profiling and optimization, ensuring high visual fidelity remains performant across target platforms. By collaborating closely with cross-discipline teams, I bridge the gap between design and execution, delivering polished, end-to-end technical solutions from initial prototyping to final production.

## TECHNICAL SKILLS

- **Languages: C++ (Advanced)**, C#, Python, JavaScript, HTML, CSS.
- **Tools & Platforms:** Git, GitHub, Visual Studio, Linux, Windows.
- **Systems & Frameworks:** **Cloud Native concepts**, System-level programming, Unreal Engine, Unity, SFML, **CMake**, OpenGL.
- **Key Competencies:** Functional Testing, Debugging, System Optimization, Performance Profiling, Agile Methodologies.
- **AI & Machine Learning:** Proficient in **Game AI architectures** (Finite State Machines, Behaviour Trees); experience in *pathfinding algorithms (A)\** and heuristic optimization.

Highly interested in applying these logic-based patterns to **Automated Root Cause Analysis** and **Network Optimization**.

## EDUCATION

---

### **BSc (Hons) Computer Science (Games Programming)**

Teesside University

(Sep 2022 – Jun 2025)

- *First Class Honours*

*Modules:*

|                                    |              |
|------------------------------------|--------------|
| <i>C++</i>                         | <i>86%</i>   |
| <i>Algorithms for Games</i>        | <i>69.5%</i> |
| <i>Game Development with C++</i>   | <i>86%</i>   |
| <i>Maths for Games</i>             | <i>74.5%</i> |
| <i>Systems and Game Technology</i> | <i>75%</i>   |
| <i>Graphics Programming</i>        | <i>70%</i>   |
| <i>Game Engine Construction</i>    | <i>70%</i>   |
| <i>Journeyman</i>                  | <i>81%</i>   |
| <i>Real Time Simulation</i>        | <i>70%</i>   |
| <i>Beta Arcade</i>                 | <i>74%</i>   |
| <i>Physics Simulation</i>          | <i>83%</i>   |
| <i>AI For Games</i>                | <i>60%</i>   |

### **OCR Level 3 Cambridge Technical Extended Diploma in IT**

Darlington College

(Sep 2021 – Jun 2021)

- *Merit Merit*

### **OCR Level 3 Cambridge Technical Foundation Diploma in IT**

Darlington College

(Sep 2021 – Jun 2022)

- *Distinction Merit Merit*

## CERTIFICATIONS

---

### **English Language Certificate B2**

Hunyadi Matyas Highschool  
(Jun 2020)

### **Cyber Security Essentials Badge**

Cisco  
(2021)

### **European Computer Driving Licence**

Hunyadi Matyas Highschool  
(Sep 2021 – Jun 2022)

### **Introduction to C++ Programming & Unreal**

University of Colorado System  
(Nov 2025)

### **More C++ Programming and Unreal**

University of Colorado System  
(Nov 2025)